

NACOTE CREEK, NJ, USA

WMO: 998002

Lat: 39.5347N

Lon: 74.4636W

Elev: 39

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	15.9	-4.2	4.4	15.4	0.9	5.8	21.0	18.9	44.1	17.8	45.0	6.1	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	91.1	74.9	88.0	73.6	85.2	72.9	77.6	86.1	76.5	84.0	75.3	81.7	7.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	133.1	81.1	74.3	128.5	80.3	73.2	123.5	79.4	41.0	86.2	39.9	83.9	38.7	81.6	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	16.8	14.9	8.4	94.7	5.8	2.6	4.2	96.6	0.8	98.1	-2.5	99.6	-6.8	101.5		
			6.9	79.9	5.4	0.7	3.0	80.4	-0.2	80.8	-3.2	81.2	-7.1	81.7		
						</										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	31.6	35.2	41.3	52.7	62.6	70.2	74.1	66.7	57.1	46.2	40.3
		DBStd	16.35	9.07	8.90	8.26	7.87	7.48	6.64	6.32	7.49	7.31	7.34	8.66
		HDD50	1852	572	421	290	53	2	0	2	2	21	164	325
		HDD65	4855	1035	834	734	383	139	24	12	36	65	262	564
		CDD50	3392	2	8	21	134	391	606	748	665	502	240	51
		CDD65	916	0	0	0	13	63	180	295	234	115	16	0
		CDH74	6896	0	0	1	127	467	1392	2459	1718	649	83	0
		CDH80	2174	0	0	0	43	118	506	857	478	155	17	0
(14)	Wind	WSAvg	6.5	6.5	6.9	7.5	8.1	7.2	6.2	5.8	5.3	6.0	6.3	6.0
(15)	Precipitation	PrecAvg	45.0	3.8	2.9	4.5	3.4	3.4	3.3	4.1	4.7	3.7	3.3	4.3
		PrecMax	58.4	7.9	5.8	8.3	6.4	8.4	6.8	7.7	8.8	6.6	8.8	8.8
		PrecMin	33.9	1.4	0.7	0.5	1.5	1.3	1.2	1.3	1.7	0.7	0.6	1.7
		PrecStd	5.4	1.5	1.3	2.0	1.3	1.6	1.5	1.5	2.0	1.9	2.1	1.6
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.0	66.3	71.4	86.3	87.5	94.7	94.6	90.4	89.7	81.4	69.9
			MCWB	52.4	56.8	58.9	63.5	70.9	75.8	76.2	73.4	73.8	67.6	59.5
		2%	DB	54.8	57.7	63.6	76.5	83.6	90.6	91.7	88.0	84.3	75.7	65.8
			MCWB	51.3	51.2	53.2	61.1	70.2	74.8	75.9	73.6	72.9	65.3	58.8
		5%	DB	50.9	52.7	58.2	69.8	79.9	86.0	88.1	85.4	80.6	71.6	61.9
			MCWB	47.8	47.5	50.9	58.8	68.1	73.4	74.5	72.9	70.2	62.2	57.2
		10%	DB	46.4	48.3	54.5	65.1	75.3	82.0	84.9	82.5	77.4	68.8	59.2
			MCWB	42.1	43.9	49.2	55.7	65.5	70.3	73.8	71.3	69.8	63.1	55.6
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	58.1	60.1	67.1	73.9	78.1	79.6	78.4	77.3	71.5	65.3
			MCDB	57.4	63.9	66.7	78.2	84.3	89.8	90.7	85.2	84.9	77.1	67.7
		2%	WB	52.1	53.0	56.0	63.9	72.0	76.4	78.1	77.0	75.3	68.5	60.5
			MCDB	53.6	56.0	60.3	72.2	80.5	87.0	87.2	83.1	80.5	72.0	62.4
		5%	WB	48.0	48.4	52.9	61.4	70.2	74.7	76.9	75.7	73.5	66.7	58.6
			MCDB	51.1	51.6	56.8	68.9	77.4	83.6	84.7	80.8	78.1	70.1	61.0
		10%	WB	42.7	44.2	49.9	58.4	67.8	72.8	75.5	74.4	72.0	64.2	55.9
			MCDB	45.6	48.1	54.7	63.1	73.0	79.8	82.1	79.5	76.0	67.6	59.0
(35)	Mean Daily Temperature Range	MDBR	16.3	16.9	16.6	18.3	17.2	16.7	17.3	17.2	16.4	17.4	17.4	15.9
		5% DB	MCDBR	21.7	22.8	23.1	25.0	22.6	21.5	20.6	19.3	19.0	20.3	21.5
			MCWBR	18.9	16.9	15.6	12.5	11.2	8.5	8.0	7.9	9.3	12.0	15.5
		5% WB	MCDBR	21.0	21.4	19.8	21.4	19.3	20.4	17.8	15.4	15.7	15.8	16.7
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.321	0.355	0.387	0.433	0.490	0.497	0.490	0.411	0.363	0.332	0.315
		taud	2.491	2.478	2.422	2.355	2.262	2.141	2.132	2.144	2.340	2.477	2.517	2.535
		Ebn at Noon	271	286	286	282	270	254	250	249	264	267	263	263
		Edh at Noon	25	29	34	38	43	48	49	47	36	28	24	22
(44)	All-Sky Solar Radiation	RadAvg	632	922	1236	1586	1759	1929	1965	1724	1413	1011	722	561
		RadStd	56	101	101	125	208	163	96	116	129	99	66	53

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+221	+130

Nomenclature: See separate page